

Q1. Data Understanding Identify all data quality issues present in the dataset that can cause problems during data loading.

From the provided dataset, the following data quality problems are present:

- **Missing values** – Sales_Amount contains Null.
- **Invalid numeric values** – "Three Thousand" is text in a numeric column.
- **Inconsistent date formats** – Dates appear in different formats (DD-MM-YYYY and YYYY/MM/DD).
- **Repeated record** – The row (O101, C001, 4500, 12-01-2024) appears twice.
- **Header inconsistency** – Column headers are duplicated/misaligned at the top of the dataset.

Q2. Primary Key Validation

Assume order_ID is the Primary Key.

a) Is the dataset violating the Primary Key rule?

Yes, the dataset violating the primary key rules.

A Primary Key must:

- Be unique
- Not contain duplicates
- Not contain NULL values

b) Which record(s) cause this violation?

Record O101 cause this violation because it appears twice.

Q3. Missing Value Analysis Which column(s) contain missing values?

a) List the affected records

Column of Sales_Amount contain missing values.

Order_ID	Customer_ID	Sales_Amount	Order_Date
O102	C002	Null	15-01-2024

b) Explain why loading these records without handling missing values is risky

Loading records with missing numeric values is risky because:

- If the column is defined as NOT NULL, the insert will fail.
- Financial calculations (totals, revenue reports) will be incorrect.
- Aggregations like SUM() may produce misleading results.
- It may indicate incomplete transactional data.

Q4. Data Type Validation Identify records where Sales_Amount violates expected data type rules.

Sales_Amount should be numeric (DECIMAL).

a) Which record(s) will fail numeric validation?

Order_ID	Sales_Amount
O104	Three Thousand

b) What would happen if this dataset is loaded into a SQL table with as DECIMAL

- ☐ Insert will fail with conversion error
- ☐ Entire batch load may stop (depending on DB settings)
- ☐ Transaction rollback may occur

Q5. Date Format Consistency The column has multiple formats.

a) List all date formats present in the dataset

DD-MM-YYYY, YYYY/MM/DD

b) Why is this a problem during data loading?

- ☐ Databases expect consistent date formats.
- ☐ Parsing errors during bulk load.
- ☐ May lead to incorrect date storage.

Q6. Load Readiness Decision Based on the dataset condition:

a) Should this dataset be loaded directly into the database? (Yes/No)

NO

b) Justify your answer with at least three reasons

- ☐ Missing Sales_Amount value
- ☐ Invalid numeric value ("Three Thousand")
- ☐ Inconsistent date formats

- ☐ Duplicate record

Q7. Pre-Load Validation Checklist

List the exact pre-load validation checks you would perform on this dataset before loading

- **Data Type Checks:** Confirm that numerical, string, boolean, or date fields match the target schema.
- **Schema Validation:** Verify that the number of columns, column names, and data types match the destination database schema.
- **Null Value/Presence Checks:** Ensure that required fields do not contain null or missing values, which can break downstream applications.
- **Uniqueness Checks:** Confirm that primary keys or unique identifiers (e.g., ID, email) do not have duplicates.
- **Format/Pattern Checks:** Validate that strings follow expected patterns, such as email addresses, postal codes, or date formats (e.g., YYYY-MM-DD).
- **Referential Integrity Check:** Confirm that foreign keys in the source data correspond to existing keys in target dimension tables.
- **Consistency Check:** Ensure data is logically consistent across related fields (e.g., Ship Date must be after Order Date).

Q8. Cleaning Strategy Describe the step-by-step cleaning actions required to make this dataset load-ready

To make the dataset load-ready, follow these simple steps:

1. **Fix headers** - Keep only one correct header row.
2. **Remove duplicate Order_ID** - Delete one of the duplicate O101 records.
3. **Handle missing values** - Fix or remove the record where Sales_Amount is NULL.
4. **Correct invalid data types** - Change "Three Thousand" to 3000.
5. **Standardize date format** - Convert all dates to one format (e.g., YYYY-MM-DD).
6. **Check numeric rules** - Ensure Sales_Amount is numeric and greater than 0.
7. **Final validation check** - Recheck for duplicates, NULLs, and format errors.

Q9. Loading Strategy Selection Assume this dataset represents daily sales data.

a) Should a Full Load or Incremental Load be used?

Incremental Load

b) Justify your choice.

Use Incremental Load because daily sales data continuously grows, and only new records need to be added, making it faster, more efficient, and less resource-intensive than a full load.